

Conditional Probability

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

A bag contains 5 red and 3 blue counters. One red counter is taken out and not replaced.
How many counters are left in the bag?

2 [2 marks]

A bag contains 5 red and 3 blue counters. One red counter is taken out and not replaced.
Work out the probability that the next counter taken is also red.

3 [2 marks]

Ten cards numbered 1 to 10 are shuffled. One card is taken and **not** replaced; it is the 3.
Work out the probability that the next card taken is even.

4 [2 marks]

The table shows information about 40 students.

	Likes science	Does not like science	Total
Likes maths	15	9	24
Does not like maths	6	10	16

A student who likes maths is chosen at random. Work out the probability that they also like science.

5 [3 marks]

A bag contains 4 green and 5 yellow counters. Two are taken without replacement.
Work out the probability that both are green.

6 [3 marks]

A bag contains 4 green and 5 yellow counters. Two are taken without replacement.
The first counter is green. Work out the probability that the second is yellow.

7 [3 marks]

In a group of 50 people, 30 are male. 12 of the males and 8 of the females wear glasses.
(a) A male is chosen at random. Work out the probability that he wears glasses.
(b) A female is chosen at random. Work out the probability that she wears glasses.

8 [3 marks]

Two cards are dealt from a shuffled pack of 52 playing cards, without replacement.
Work out the probability that both are hearts. (There are 13 hearts in a pack.)

9 [4 marks]

80 students were asked how they travel to school. The table is incomplete.

	Bus	Walk	Total
Year 10	27		45
Year 11			
Total	50		80

A student who takes the bus is chosen at random. Work out the probability that the student is in Year 11.

10 [4 marks]

A bag contains 3 red, 4 blue and 5 green counters. Two counters are taken at random without replacement.
(a) Work out the probability that both are red.
(b) The first counter is blue. Work out the probability that the second is green.

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

$P(A) = 0.4$ and $P(A \text{ and } B) = 0.1$

Work out $P(B | A)$.

2

[2 marks]

A bag contains 7 red and 5 blue counters. Two are taken without replacement, and the first is red.

Write down the probability that the second is also red.

3

[3 marks]

A bag contains 6 red and 6 blue counters. Two are taken without replacement.

Work out the probability that both are red.

4

[3 marks]

Using the same bag (6 red, 6 blue, two taken without replacement),
work out the probability that exactly one counter is red.

5

[4 marks]

In a survey of 60 people, 35 own a dog, 28 own a cat and 12 own both.

(a) A cat owner is chosen at random. Work out the probability they also own a dog.

(b) A dog owner is chosen at random. Work out the probability they also own a cat.

(c) Work out the probability that a randomly chosen person owns neither.

6

[4 marks]

5% of a population has a certain condition.

A test gives a positive result for 90% of people who have the condition, and also gives a positive result for 8% of people who do not have it.

(a) Work out the probability that a randomly chosen person tests positive.

(b) A person tests positive. Work out the probability that they actually have the condition, to 3 significant figures.

7

[4 marks]

A bag contains 5 red and 3 blue counters. Three are taken without replacement.

Work out the probability that all three are red.

8

[4 marks]

$P(A) = 0.6$, $P(B | A) = 0.3$ and $P(B | \text{not } A) = 0.5$

(a) Work out $P(B)$.

(b) Work out $P(A | B)$.

9

[4 marks]

A bag contains 5 white counters and n black counters.

Two counters are taken without replacement. The probability that both are white is $\frac{5}{14}$

Work out the value of n .

A box contains 4 milk chocolates and 6 dark chocolates. Two are taken at random without replacement.

- (a) Work out the probability that both are milk.
- (b) Work out the probability that the second chocolate is dark.
- (c) Given that the second chocolate is dark, work out the probability that the first was milk.

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